

VORA: streaming voice Q&A on CPU (ASR → RAG → LLM → TTS)

Free, open-source models, CPU only, English and Mandarin. Every number below comes from results/*.json (final run 2026-10-06, Apple M2, 8 cores, test splits read once with --final). Full design notes and every earlier table: docs/appendix.md.

1. Architecture

Browser mic (16 kHz PCM16, 100 ms frames, AudioWorklet) → WebSocket → **streaming ASR** (sherpa-onnx: zipformer-20M en int8 41.6 MB, CTC small zh int8 25 MB) → partial text to the UI; stable partials prefetch retrieval → **hybrid retrieval** (int8 bge-small en/zh per language, FAISS + sentence passages + BM25, calibrated refusal threshold) → **Qwen2.5-0.5B Q4_K_M** (llama.cpp) token stream → **answer guard** (negation, figures, refusals answered from the retrieved text) → chunker (1 word, 3 words, then sentences) → **Piper VITS TTS** (en LJSpeech int8, zh huayan) → 16 kHz audio back on the same socket.

Each user gets one socket; ASR, retrieval, LLM and TTS run on separate executors with a bounded output queue, so a slow client only blocks its own TTS. Speaking over the answer cancels it (barge-in). The browser client is plain ES modules with a state machine covering every state (loading, busy, mic errors, close codes, iOS interruption) and was checked at 10 screen sizes, light and dark.

2. Gates (brief targets)

Gate	Target	Result	Measured
G1 latency, synthetic en audio	p50 ≤1500, p90 ≤1800 ms	PASS	p50 1006 / p90 1438 ms
G1r latency, real voices (MInDS-14)	answered p50 ≤1500, p90 ≤1800 ms	PASS	p90: en-US 1337, en-GB 1422, en-AU 1725, zh 1507 ms (§3)
G2 TTS first chunk	p95 ≤200 ms	PASS	en 70 / zh 89 ms
G3 ASR+RAG+TTS memory	≤500 MB USS	PASS	305 MB (macOS); 465 MB in the Linux container
G4 ASR accuracy, clean read speech	WER/CER ≤15%	PASS	en LibriSpeech 8.4%, zh AISHELL 5.8%
G5 retrieval + faithfulness	top-3 ≥80%, faithful ≥95%	FAIL	top-3 96%, faithful 89% of 100 (blind 81.7%)
G6 two users at once	each p50 ≤2× single	PASS	×1.36 (1034 / 1215 vs 891 ms)
G7 Docker + Pi-class proof	arm64, amd64, limited cores, Pi-class CPU	UNVERIFIED	3 of 4 done; Pi-class run refused by AWS free plan
G8 permissive licences	no unknown licences	FAIL	zh voice huayan: licence unknown (kept by owner ruling)

Latency = speech end to first audio of the answer, in-process through the real models. LLM memory is reported separately: 653 MB.

3. Real human speech (third-party, not self-authored)

Questions and audio come from MInDS-14 (PolyAI, CC-BY-4.0): real callers asking a bank about 14 intents, recorded over the phone at 8 kHz. They are answered from a separate demo bank knowledge base written from the intent names only, before any utterance was read. Accuracy is on 100 test clips per cell; latency on 42 fixed clips per language (14 per English accent).

Scene	ASR error	Top-3 (ASR text / exact text)	Faithful (ASR / exact text)	Answered p50 / p90 ms (n)
en-US phone, 8 kHz	WER 44.0%	90% / 98%	52% / 67%	1061 / 1337 (13 answered, 1 refused)
en-GB phone	WER 49.1%	74% / 93%	42% / 57%	1178 / 1422 (10, 3)
en-AU phone	WER 42.0%	76% / 96%	43% / 56%	1034 / 1725 (11, 3)
zh-CN phone	CER 21.5%	83% / 92%	59% / 60%	1006 / 1507 (28, 14)

Latency counts answered turns, as the gate defines it; refused turns get a fixed short reply and are 7–33% of the latency clips (zh highest). Speech end is cross-checked against the Silero VAD (median gap 68 ms on en-US).

Noise and room effects on the same speakers (top-3 on ASR text):

Scene	en-US WER / top-3	zh-CN CER / top-3
clean phone	44.0% / 90%	21.5% / 83%
ambient noise 10 dB	45.4% / 87%	22.9% / 83%
ambient noise 5 dB	53.8% / 74%	25.6% / 77%
babble 10 dB	77.5% / 70%	56.2% / 81%
reverb 0.6 s	92.3% / 8%	39.5% / 56%
quiet (−30 dB) / loud (+12 dB, clipped)	41.9% / 88%, 44.7% / 80%	21.7% / 84%, 21.9% / 82%

Read speech (WER/CER, clean → ambient 10 dB): LibriSpeech clean 5.9% → 7.4% (phone codec 8.3%), LibriSpeech other 12.8% → 16.0%, FLEURS en 25.9% → 28.7%, AISHELL 5.2% → 5.8%, FLEURS zh 12.3% → 14.6%.

Reading it. Phone speech is hard for 16 kHz models at the 50 MB size limit (WER 42–49%). Retrieval still finds the right answer most of the time because a few key words are enough: the exact-transcript column shows the gap that ASR costs. Reverberation is the failure case: the English model collapses (8% top-3), so the system refuses rather than guessing (77% refused). Noise reduction (GTCRN) was tested and made accuracy worse, so it is not in the live path.

4. What changed in this round

- Retrieval: a word-boundary bug dropped synonyms for Latin words written next to Chinese (“5g的wifi”); fixed. Added per-language int8 embedders, sentence-level passages, normalised BM25 and per-language refusal thresholds tuned on dev only. Blind sets are frozen by sha256 and used once.
- English voice: LJSpeech (public domain, trained from scratch) replaced lessac; first chunk p95 70 ms.
- Memory in Linux: int8 embedders without fastembed, numpy resampler: 560 → 465 MB USS.
- Automatic gain control before ASR fixes quiet speakers (LibriSpeech at −30 dB: WER 42.9% → 5.6%, dev split) and costs at most 1.3 points elsewhere.
- Deployment: Docker images for arm64 and amd64, a demo on AWS (below), and an access key for the WebSocket.

5. Deployment

`docker build -f docker/Dockerfile -t vora .` fetches pinned models and builds the index. Then `docker run -p 8000:8000 vora` and open `http://localhost:8000`. For other devices, use HTTPS (`scripts/make_cert.sh`), because browsers allow the microphone only on secure pages. Set `VORA_ACCESS_KEY` when the port is reachable from outside.

Proof	Result
arm64 image (native)	built, ready in 84 s, WebSocket smoke test passes in English, Chinese and with a real voice
amd64 image (emulated)	built 1.06 GB, boots, smoke test passes; latency under emulation not meaningful
Pi 4 shape (—cpus=4 —memory=1g, fast M2 cores)	first audio p50 979 / p95 2811 ms; Linux USS 465 MB
AWS t4g.small demo (2 Graviton2 vCPU, 2 GB)	HTTPS on 443, access key required; 3.8–5.6 s per answer: works, does not meet G1
Pi-class CPU (a1.xlarge, Cortex-A72)	not run: AWS free plan refused the instance type
Jetson	not run: no hardware. Same arm64 image, CPU only

6. Limitations (stated, not hidden)

- G5 faithfulness 89%, not 95%.** The misses are retrieval near-misses and the 0.5B model paraphrasing wrongly; off-topic refusal on the blind set is 60% (6 of 10). On real phone speech faithfulness is far lower, because the ASR text is wrong before retrieval starts.

- **Raspberry Pi and Jetson unmeasured.** The closest evidence is the 4-core Docker run on M2 cores, which is optimistic. A Pi 4 core is roughly 1.5× slower than the A72 in a1 and far slower than an M2.
- **Chinese voice licence unknown** (huayan). An Apache-2.0 swap exists (aishell3, 40 MB), but it is over the 30 MB TTS limit.
- **Phone speech and reverb.** No ≤ 50 MB streaming English model was found that does better; a second pass model (27.6 MB) would break the per-model size limit together with the first.
- **Testing.** Real microphones, Firefox, and physical phones were not tested automatically; a manual checklist is in docs/demo-script.md. The VORA Box question sets are self-authored and labelled as such; the real-voice results above are not.
- **Cantonese** is not supported.